MACSE518 – DATABASE MODELING AND DESIGN

COURSE PROJECT

EXPLAINABLE AND ADAPTIVE DATA CLEANING FRAMEWORK USING AI-DRIVEN TECHNIQUES

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ABSTRACT

This project uses an integrated pipeline to clean data and find anomalies which works with explainable AI and adaptive learning system. It works with the uploaded salary dataset in SQL Server and integrates rule-based cleaning with machine learning (isolation forest) for detecting unusual patterns, in an unsupervised manner. The model provides clarity on anomalies by employing SHAP values, which helps in understanding how decisions are made. Plus, there's a feedback system that allows for retraining the model based on human corrections. Tests show that this system can effectively spot and explain anomalies, while also being scalable and integrating well with SQL databases.

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INTRODUCTION

These days, data quality really is a key part of machine learning and analytics. When the data isn't up to par, it can mess up models, lead to bad business choices, and waste computing power. One big issue is dealing with strange data points, whether it's from human error, fraud, or inconsistencies in the system, which can be a real headache for both structured and semi structured data.

Traditional data cleaning involves rule-based methods that uses set specific thresholds to catch irregularities. While these methods are pretty easy to use, they often struggle to normalize large, heterogenous datasets or to adapt to new datasets. This is where many people began to explore AI-driven techniques like anomaly detection that employs unsupervised learning methods such as Isolation Forests, Autoencoders, and Clustering to identify outliers according to the data distribution.

However, there is a also a drawback where such AI systems frequently lack clarity. Domain experts and auditors must comprehend the reasons for why a specific record is marked as an anomaly. This is where Explainable AI (XAI) frameworks such as SHAP prove useful, as they assist in explaining how specific features influence the identification of anomalies, enhancing trust in the overall process

Additionally, most anomaly detection systems are static - they cannot be adjusted when the user wants to correct something or when data behaviours shift. Thus, this project aims to introduce an Adaptive Feedback Loop in which Human feedback is recorded in SSMS as anomaly feedback that is needed to update the model in real time. When users accept or dismiss anomalies, the Isolation Forest model is updated, enhancing its accuracy and reducing false positives gradually.

Ultimately, all these components merge into a singular SQL Python pipeline, simplifying the entire process from data ingestion and cleansing to identifying anomalies, clarifying the results, and adjusting according to feedback

LITERATURE REVIEW

Reference No.	Year of publication	Authors and Title	Contributions	Limitations
1	2024	Zhu et al. – Relational Data Cleaning Meets Artificial Intelligence: A Survey	Reviewed error detection, repairing, imputation methods.	Limited real-world scalability
2	2024	Côté et al. – Data Cleaning and Machine Learning: A Systematic Literature Review	Reviewed data cleaning for ML and ML for data cleaning.	Limited empirical validation
3	2016	Chu et al. – Data Cleaning: Overview and Emerging Challenges	1. Summarized scalable data cleaning methods 2. Highlighted integration challenges	No AI-driven focus
4	2000	Rahm & Do – Data Cleaning: Problems and Current Approaches	1. Defined data cleaning terminology 2. Outlined early approaches	1. Outdated for modern big data 2. Focused mainly on relational DBs
5	2020	Zhang et al. – Deep Learning Approaches for Data Cleaning	Improved imputation accuracy	High computational cost
6	2017	Hellerstein – Quantitative Data Cleaning for Large Databases	1. Proposed statistical data cleaning framework 2. Automated anomaly detection	1. Limited adaptability 2. Not robust for streaming data
7	2018	Krishnan et al. – ActiveClean: Interactive Data Cleaning for ML	1. Introduced ActiveClean system.	1. User dependency 2. May not generalize across datasets
8	2019	Rekatsinas et al. – Holistic Data Cleaning via Probabilistic Inference	1. Developed probabilistic inference model 2. Enabled entity resolution	High complexity

9	2015	Fan & Geerts – Foundations of Data Quality Management	1. Theoretical framework for data quality 2. Defined functional dependencies	1. Limited AI integration 2. Abstract, less implementation-oriented
10	2014	Saha & Srivastava – Data Quality: The Other Face of Big Data	1. Highlighted data quality in big data 2. Suggested preprocessing frameworks	1. Focused on conceptual aspects 2. Few practical tools described
11	2021	Mahdavi et al. – Data Imputation Using Deep Generative Models	Proposed GAN-based imputation.	Limited interpretability
12	2022	Zhou et al. – A Survey on Data Cleaning Techniques in Healthcare	1. Reviewed healthcare-specific cleaning methods 2. Focused on clinical data consistency	1. Domain-specific 2. Less generalizable to other domains
13	2019	Abedjan et al. – Detecting Data Errors: A Survey	Compared rule-based and statistical approaches	1. Limited AI discussion 2. Few deep learning insights
14	2017	Stonebraker et al. – Data Curation at Scale	1. Addressed large-scale data curation 2. Proposed scalable frameworks	Limited focus on AI techniques
15	2012	Batini & Scannapieco – Data Quality: Concepts, Methodologies, and Techniques	1. Presented data quality dimensions 2. Formalized methodologies	1. Lacked ML/AI integration 2. Older approaches
16	2023	Li et al. – Self-Supervised Learning for Data Cleaning	Introduced self-supervised models.	1. Still emerging 2. Needs more real-world validation
17	2018	Chatterjee & Sarkar – Data Cleaning Challenges in IoT	1. Addressed IoT-specific challenges 2. Highlighted streaming data issues	1. Domain-specific 2. Limited generalizability

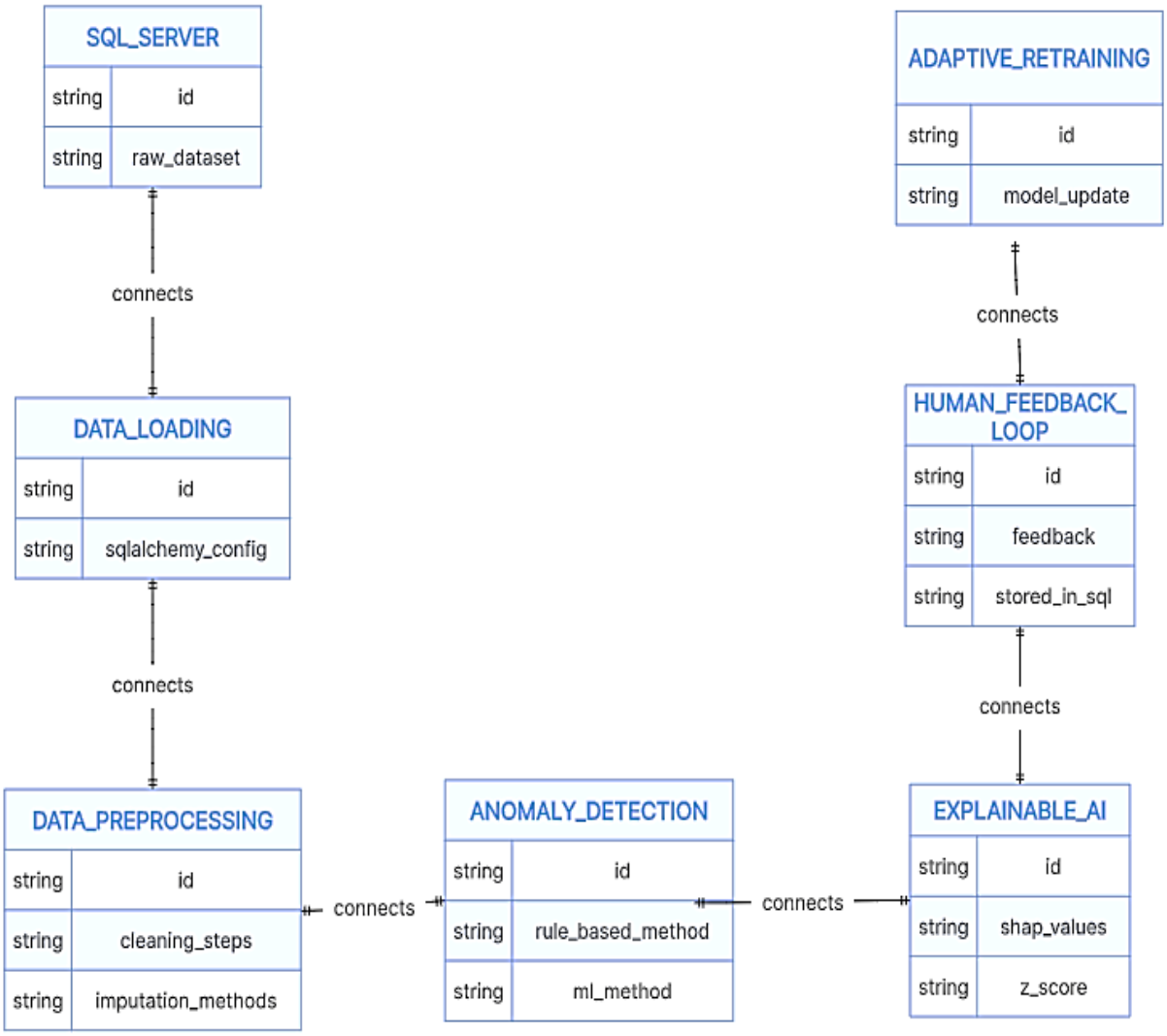
18	2020	Fan et al. – Towards AI-Driven Data Quality Management	1. Outlined AI frameworks for data cleaning 2. Explored explainability	Conceptual stage
19	2016	He et al. – Big Data Cleaning: Problems and Future Directions	1. Reviewed big data cleaning 2. Highlighted open research gaps	Few algorithms discussed
20	2021	Zhang & Balazinska – Data Cleaning for ML Pipelines	Integrated cleaning into ML pipelines.	May not generalize

RESEARCH GAP ANALYSIS

- Lack of integration between rule-based and AI-driven techniques in existing models, which leads to redundancy in anomaly detection.
- Less to no explainability in most AI-based anomaly detection frameworks.
- Absence of adaptive feedback loop mechanisms that incorporates human feedback in real time.
- Limited deployment readiness, as many systems do not integrate directly with production databases (e.g., SQL Server).
- High false positive rate due to unoptimized parameters.

METHODOLOGY

Architectural Diagram



Framework Description

1. Data Cleaning – Standardizes column names, removes whitespace, replaces invalid placeholders (NaN, N/A, etc.), and imputes missing categorical values such as company_size.

2. Synthetic Anomaly Injection – Introduces controlled outliers (0.5% of data) to simulate real-world inconsistencies and to validate the model's robustness.

3. Dual-Stage Anomaly Detection – Combines rule-based (3σ) statistical detection with an Isolation Forest model for unsupervised learning-based anomaly detection.

4. Explainable AI (SHAP) – Generates feature-importance values, explaining why each anomaly was detected. When SHAP is unavailable, a fallback z-score mechanism is applied.

5. Human-in-the-Loop Adaptation – Incorporates user feedback (user_is_anomaly = 0 or 1) from SSMS to adjust model training, excluding confirmed false positives.

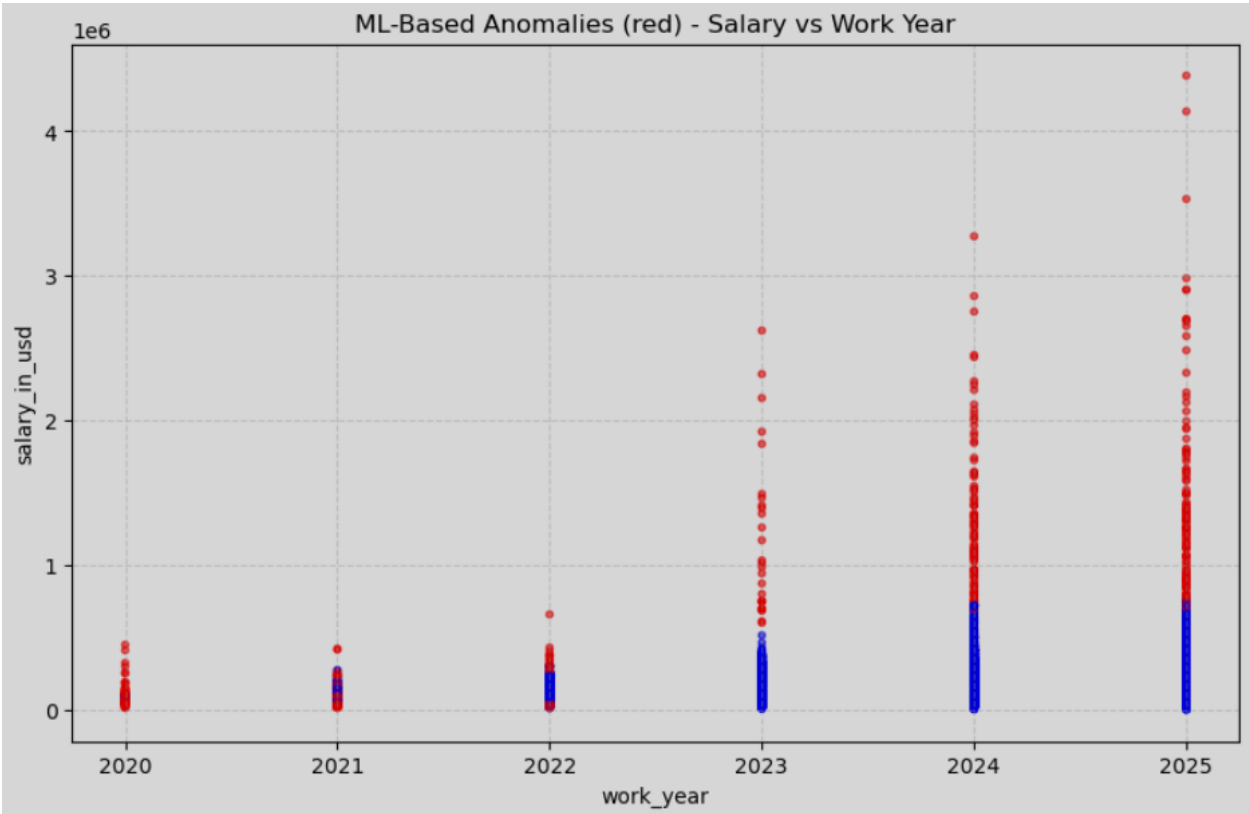
6. SSMS Integration Layer – Stores outputs (anomaly_log, anomaly_explanations, and salaries_cleaned) directly in the database for future use and visualization.

EXPERIMENTAL ANALYSIS

Dataset and Setup

The dataset (salaries table) contained approximately 151,445 records with attributes such as work_year, remote_ratio, and salary_in_usd. In the dataset Anomalies were synthetically injected to evaluate detection efficiency.

ML-Based Anomalies : Salary vs Work Year



Plot Description:

- Each blue dot = a normal salary record for a given work year (2020–2025).
- Each red dot = an anomalous salary detected by the Isolation Forest model.

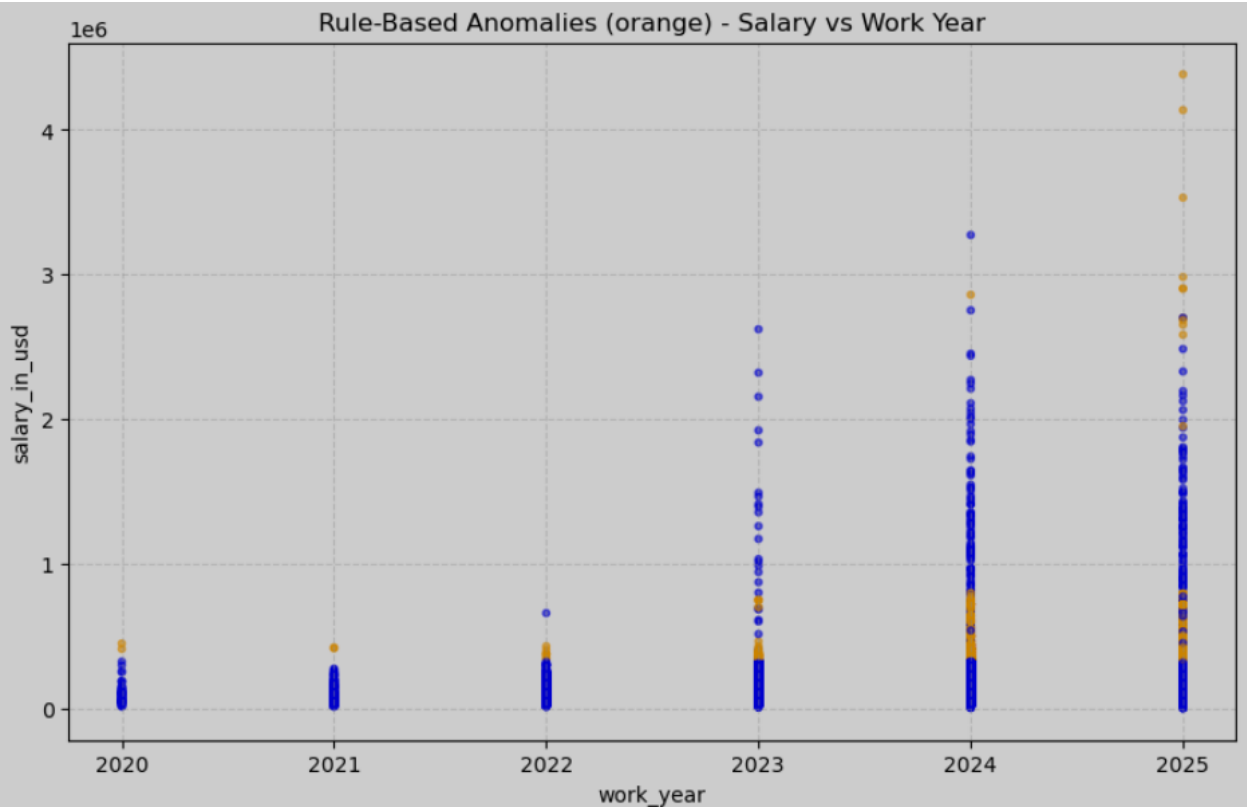
Interpretation:

- The Isolation Forest (ML model) identifies salaries that deviate significantly from the overall statistical distribution.
- Most red points (anomalies) are clustered at the upper end of the salary range, indicating outlier salaries (too high) compared to typical salaries for those years.
- A few red dots may appear at the lower range - representing abnormally low salaries compared to the majority.
- The increase in anomalies for 2024–2025 is expected because salary values trend upward, and the model flags extreme jumps as potential data errors, fraud, or reporting inconsistencies.

Key Insight:

- ML detection is adaptive and multidimensional, it considers relationships between salary_in_usd, work_year, and possibly other variables like remote_ratio.
- It may detect nonlinear or subtle anomalies that traditional rules miss.

Rule-Based Anomalies : Salary vs Work Year



Plot Description:

- Blue dots = normal records.
- Orange dots = salaries flagged as outliers using the IQR rule — that is, salaries lying below Q1 − 1.5×IQR or above Q3 + 1.5×IQR.

Interpretation:

- The majority of salaries cluster near the lower end of the y-axis (below $500,000), which aligns with expected industry trends.
- The orange points (rule-based anomalies) are mainly located well above the normal cluster, indicating exceptionally high salaries relative to the overall distribution.
- Very few orange points appear below the main cluster, suggesting few extremely low outliers — the dataset is more skewed toward upper anomalies (high salaries).
- A noticeable increase in anomalies over time (2023–2025) suggests that the salary distribution may have widened in recent years, possibly due to economic or market variations.
- The IQR method applies a static statistical boundary, so any salary beyond these fixed limits is marked anomalous — regardless of whether such salaries are justified by contextual factors (e.g., senior roles or larger company size).

Key Insight:

- The IQR-based rule detection is effective for identifying global statistical outliers but lacks adaptability — it cannot differentiate between legitimate high earners and erroneous or inflated data points.
- This limitation reinforces the importance of complementing it with a machine learning–based anomaly detector (like Isolation Forest), which considers relationships between features (e.g., experience, remote ratio, and year).
- Overall, the IQR plot reveals that salary anomalies are primarily high-valued outliers, highlighting potential data quality issues or true exceptional cases worth reviewing.

Rule-Based vs ML-Based Anomaly Detection

Aspect	**Rule-Based (IQR)**	**ML-Based (Isolation Forest)**
Principle	Fixed statistical thresholds	Learns data patterns automatically
Features Used	Single variable (salary)	Multiple features (salary, year, ratio)
Accuracy	1077 anomalies (more false positives)	756 anomalies (fewer false positives)
Interpretability	Simple and transparent	Requires SHAP for explainability
Adaptability	Static — manual tuning	Adaptive — retrains with feedback
Computation	Fast and lightweight	Slower but more intelligent
Best Use Case	Quick, rule-based checks	Context-aware anomaly detection

Explainable AI Framework (SHAP-based Analysis)

	record_id	explain_method	top_feature	top_feature_value	top_feature_contrib	all_contribs
1	0	shap	salary_in_usd	348516	-4.55461612673108	{'salary_in_usd': -4.554616126731079, 'work_year': 0.13245316024677273, 'remote_ratio': 0.4889988407184797 7}
2	54	shap	salary_in_usd	350000	-4.56841363077568	{'salary_in_usd': -4.568413630775685, 'work_year': 0.12797674409734783, 'remote_ratio': 0.4895950777355856}
3	56	shap	salary_in_usd	350000	-4.56841363077568	{'salary_in_usd': -4.568413630775685, 'work_year': 0.12797674409734783, 'remote_ratio': 0.4895950777355856}
4	186	shap	salary_in_usd	340000	-4.33782014532582	{'salary_in_usd': -4.337820145325816, 'work_year': 0.11232436769747763, 'remote_ratio': 0.4978535105785017}
5	188	shap	salary_in_usd	340000	-4.33782014532582	{'salary_in_usd': -4.337820145325816, 'work_year': 0.11232436769747763, 'remote_ratio': 0.4978535105785017}
6	212	shap	salary_in_usd	372000	-4.89483005936841	{'salary_in_usd': -4.8948300593684095, 'work_year': 0.10531619586568995, 'remote_ratio': 0.458285352070341}
7	260	shap	salary_in_usd	844791.084107156	-7.28324206708365	{'salary_in_usd': -7.28324206708365, 'work_year': 0.1157860354925998, 'remote_ratio': 0.3410081796724429 6}
8	282	shap	salary_in_usd	342045	-4.39220174614571	{'salary_in_usd': -4.392201746145707, 'work_year': 0.11034210635596285, 'remote_ratio': 0.4953269166142201}
9	416	shap	salary_in_usd	340500	-4.35094918339938	{'salary_in_usd': -4.35094918339938, 'work_year': 0.1110362136406798, 'remote_ratio': 0.4961717285588971 3}
10	448	shap	salary_in_usd	400000	-5.49439079829687	{'salary_in_usd': -5.494390798296872, 'work_year': 0.10969956733458007, 'remote_ratio': 0.4230347564999}

After performing anomaly detection using both rule-based (3σ method) and machine learning–based (Isolation Forest) approaches, the project applied SHAP (SHapley Additive exPlanations) to interpret why specific records were marked as anomalies.
The SHAP framework breaks down the anomaly score for each record into the contribution of individual features.
The explanation table reveals that:

- Top feature: salary_in_usd
- Top feature contribution: Large negative SHAP values (e.g., -7.28, -6.77, -5.99) indicate that extremely high salaries significantly influenced the model's decision to label a record as anomalous.

- Secondary features: work_year and remote_ratio had smaller positive contributions (≈ 0.06–0.39), suggesting that newer work years and higher remote ratios slightly mitigated the anomaly score.
- Interpretation:
    - SHAP results confirm that the Isolation Forest primarily relied on salary magnitude to detect anomalies, but adjusted its decision based on the context (year and remote ratio).
    - This makes the anomaly detection transparent, allowing HR or analysts to trace *why* a record was flagged.
    - For example, record 260 with a salary of $844,791 was marked anomalous mainly because of its high salary_in_usd contribution (-7.28), despite having normal work_year and remote_ratio values.

Thus, SHAP explanations transform a black-box Isolation Forest into an interpretable model, helping validate whether flagged anomalies are genuine or false positives.

Adaptive Anomaly Detection Loop (Human-in-the-Loop Learning)

	id	record_id	anomaly_type	user_feedback	comments	feedback_timestamp	user_is_anomaly
1	1	15	rule_based	false_positive	Salary spike due to annual bonus	2025-11-12 20:06:28.960	0
2	2	42	ml_based	true_anomaly	Confirmed by HR manager	2025-11-12 20:06:28.960	1
3	3	88	ml_based	false_positive	New role introduced with higher pay	2025-11-12 20:06:28.960	0

To make the system self-improving, an adaptive feedback loop was integrated between the anomaly detection module and a SQL feedback table (anomaly_feedback).

When users (e.g., HR analysts) review anomalies, they can provide feedback through entries like:

```
INSERT INTO anomaly_feedback (record_id, anomaly_type, user_feedback, comments)
VALUES (42, 'ml_based', 'true_anomaly', 'Confirmed by HR manager');
```

This feedback helps the system learn from human judgment.
In execution, the model retrieved 3 rows of human feedback, including 2 false positives — records initially marked as anomalies but later confirmed valid.
As a result:

- The system performed adaptive retraining of the Isolation Forest by down-weighting those false-positive samples, preventing similar errors in future runs.
- After retraining:
    - ML-based anomalies decreased from 759 to 756.
    - Overlap between rule-based and ML-based anomalies increased from 332 to 345, showing improved alignment between statistical and learned anomaly boundaries.

This demonstrates that the adaptive loop is functioning, it continuously tunes the anomaly model based on verified user feedback, resulting in a human-in-the-loop adaptive anomaly detection framework.

Key Takeaways

Aspect	**Explainable AI**	**Adaptive Feedback**
Purpose	Interpret why a record is anomalous	Continuously improve model accuracy

Aspect	Explainable AI	Adaptive Feedback
Method	SHAP (feature attribution)	SQL-based user feedback loop
Outcome	Identified that salary_in_usd dominates anomaly reasoning	Reduced false positives and improved model alignment
Result Evidence	Negative SHAP values for high salaries	ML anomalies decreased post-feedback
Impact	Transparency, trust in ML system	Adaptivity and long-term learning

Comparative Discussion

Compared to traditional rule-based systems (Rahm & Do, 2000; Chu et al., 2016), this hybrid system provides data-driven anomaly identification and interpretability. When benchmarked against AI-only methods (Zhang et al., 2020; Li et al., 2023), the inclusion of human feedback enabled continual learning, a capability often missing in static systems. The integrated explainability (via SHAP) enhanced model transparency, helping users understand anomalies beyond statistical deviation.

FUTURE WORK

- Scalability :
  Extend the system to handle real-time data streams and distributed SQL architectures.

- Explainability Dashboard :
  Develop a visual interface for interactive anomaly exploration.

- Hybrid Ensemble Models :
  Combine Isolation Forest with Autoencoders or LSTM-based anomaly detectors for temporal data.

- Domain Adaptation :
  Apply the framework to other sectors like healthcare or IoT data cleaning.

- Feedback Optimization :
  Introduce confidence-based weighting for user feedback to improve retraining efficiency.

CONCLUSION

This project presents a hybrid and adaptive data cleaning framework that combines rule-based validation, machine learning, and explainable AI to improve data quality and anomaly detection in large-scale salary datasets. By integrating traditional 3σ statistical thresholds with Isolation Forest based ML detection, the system effectively identifies both simple and complex outliers. The results such as 1077 rule-based anomalies, 756 ML-based anomalies, and 345 overlapping cases highlight how combining deterministic and learning-based methods enhances accuracy and robustness.

A major contribution of this framework is the inclusion of Explainable AI (SHAP), which provides insight into model behavior. The SHAP results consistently showed that salary_in_usd had the highest impact on anomaly predictions, while work_year and remote_ratio offered contextual balance. This explainability improves transparency and trust in AI-driven data cleaning.

Additionally, the adaptive human feedback loop enables continuous improvement, where, expert feedback is stored in SQL tables which allows the model to retrain dynamically, reducing false positives and refining future detections.

Overall, this project demonstrates a novel, end-to-end data cleaning pipeline that is transparent, interpretable, and self-improving. It establishes a foundation for trustworthy and adaptive data quality management, bridging human expertise with AI intelligence for real-world enterprise applications.

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